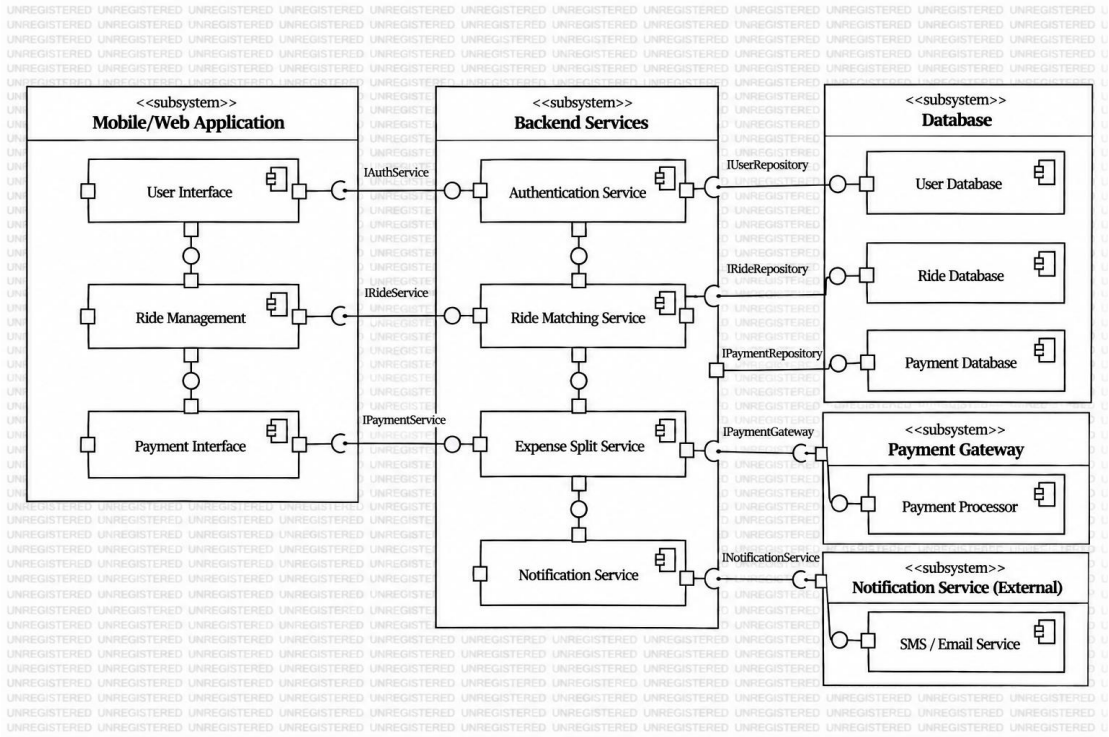
SOFTWARE ENGINEERING LAB 3
PEER-TO-PEER CARPOOLING & EXPENSE SPLITTER – UML COMPONENT DIAGRAM

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UML Component Diagram



· The system is designed for **peer-to-peer carpooling and sharing travel expenses**.
· The system has a **Mobile/Web Application** through which users interact with the system.
· The **User Interface** allows users to access different features of the application.
· The **Ride Management** component handles creating, searching, matching, and managing rides.
· The **Payment Interface** is used to handle payment-related activities in the application.
· The **Backend Services** contain the main services that process user requests and application operations.

· The **Authentication Service** manages user login and authentication.
· The **Ride Matching Service** helps match riders with suitable rides based on the required information.
· The **Expense Split Service** calculates and manages the sharing of travel expenses between users.
· The **Notification Service** sends important updates and notifications to users.
· The system uses a **Database subsystem** to store user, ride, and payment information.
· The **User Database** stores information related to registered users and their accounts.
· The **Ride Database** stores details about rides, drivers, and ride-related information.
· The **Payment Gateway** connects the system with the **Payment Processor** to process payments securely.
· The **Notification Service (External)** connects to an **SMS/Email Service** to send notifications to users.